

# Interfacial Maxwell–Wagner Relaxation Limits Zero-Bias Capacitive Readout in Ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ Capacitors

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**Abstract**—The capacitive memory window (CMW) of ferroelectric capacitors enables non-destructive readout, yet the mechanism that sets its usable bandwidth and temperature range remains unclear. We condense the full TMAP manuscript into a conference format by keeping the complete plot set and the core experimental comparison between  $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{TiN}$  and  $\text{TiN}/\text{PE-TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{WO}_x/\text{W}$  capacitors. Frequency-resolved  $C$ - $V$  spectroscopy from 1 kHz to 4 MHz, together with PUND and switching-current measurements at 300 K and 10 K, shows that the zero-bias CMW collapses as a single-Debye relaxation with  $f_c = 179$  kHz in Stack A but only  $f_c = 1.4$  kHz in the  $\text{WO}_x$ -bearing stack. At 10 K the read window falls below 1 kHz although polarization switching persists. The combined evidence identifies an interface-limited Maxwell–Wagner process: polarization stores the bit, whereas interfacial conductivity determines whether and how fast the states remain capacitively distinguishable.

**Index Terms**—Ferroelectric capacitor, hafnium zirconium oxide, capacitive memory window, non-destructive readout, Maxwell–Wagner relaxation, IEDM-style conference manuscript.

## I. INTRODUCTION

Ferroelectricity in doped  $\text{HfO}_2$  and especially in  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  has reopened the path to scalable ferroelectric memory compatible with CMOS and BEOL processing [1]–[3]. In a conventional ferroelectric capacitor, however, reading the stored state is destructive because the switched charge  $2P_r$  must be sensed by driving the device through its coercive field. A capacitive read avoids that penalty. The two remanent polarization states occupy different small-signal permittivity branches, so a sub-coercive capacitance measurement can distinguish the stored bit without forcing a rewrite [4]. That possibility is attractive for read-dominated memory operation and for analog or in-memory computing schemes that repeatedly interrogate the same stored state.

Prior work established that a capacitive memory window can be opened and enlarged by interface engineering, and that non-destructive readout can be demonstrated in BEOL-compatible ferroelectric capacitors and array-relevant circuits [5], [6]. What those reports do not resolve is the physics that governs the window itself. A large remanent polarization is necessary to define two stored states, but it is not sufficient to guarantee a useful zero-bias CMW: in an ideally symmetric stack, the two  $C$ - $V$  branches would still intersect at zero bias. The practical questions are therefore sharper than simply

asking whether ferroelectricity is present. What determines the usable read bandwidth? Why does the zero-bias window vary between stacks? And will the same readout survive across temperature?

The full TMAP paper addresses those questions by testing two competing pictures. In a ferroelectric-limited picture, the CMW should follow the intrinsic dielectric and switching dynamics of the  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  layer and therefore persist on the ferroelectric timescale. In an interface-limited picture, the window should relax with the impedance of the interfacial region and should change strongly with stack design and temperature. This conference version retains the decisive evidence for the latter view. It develops the argument around one statement: polarization stores the bit, but the interface sets whether and how fast the two states remain capacitively distinguishable.

## II. EXPERIMENTAL METHODS

Two metal–ferroelectric–metal capacitors were compared. Stack A was  $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{TiN}$  on Si. Stack B was  $\text{TiN}/\text{PE-TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{WO}_x/\text{W}$  on Si, where the deliberate  $\text{WO}_x$  layer modifies the bottom interface and provides a second stack with a different interfacial impedance. Grazing-incidence XRD was used to confirm crystallization of the ferroelectric films, while the electrical study focused on the relation between the two remanent branches of the small-signal dielectric response.

Frequency-resolved  $C$ - $V$  measurements were performed in parallel  $C_p$ - $R_p$  mode with a 50 mV rms signal at 61 logarithmically spaced frequencies from 1 kHz to 4 MHz. The dc bias was swept  $-2 \rightarrow +2 \rightarrow -2$  V so that the two polarization branches were collected under identical conditions. For each frequency, the zero-bias capacitive memory window was obtained from the branch separation,

$$\text{CMW}_\xi(f) = \xi_\downarrow(0, f) - \xi_\uparrow(0, f), \quad (1)$$

where  $\xi_\uparrow$  and  $\xi_\downarrow$  denote the branches reached after opposite preset states. The crossing field  $E_{\text{cross}}$  was taken from the bias at which the two branches became capacitively indistinguishable.

To test whether changes in the window simply mirrored changes in polarization, the capacitive data were paired with PUND and switching-current measurements. The same devices were measured at 300 K and 10 K so that the zero-bias readout

and the switching response could be compared on the same temperature axis. High-frequency interpretation was limited to the range below the series-resistance-dominated region identified from the measured roll-off and standard dielectric-analysis practice [7], [8].

### III. RESULTS AND DISCUSSION

The starting point is the contrast between the two stacks at low frequency. At 1 kHz, both devices show the butterfly-shaped permittivity characteristic of ferroelectric  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ , and both reach a similar peak relative permittivity of about 62. Yet the zero-bias state separation is different: Stack A gives  $\text{CMW}_\xi \approx 5.2$ , whereas Stack B gives only  $\approx 2.3$  [Fig. 2]. This immediately suggests that the zero-bias read margin is not set only by the ferroelectric dielectric amplitude. As the measurement frequency increases, the branch separation collapses while a substantial background permittivity remains. The disappearing quantity is therefore the state discrimination itself, not the dielectric response as a whole.

The frequency dependence of that state discrimination is captured by a single-Debye form,

$$\text{CMW}(f) = \frac{\text{CMW}_0}{1 + (f/f_c)^2}, \quad (2)$$

which directly isolates the relaxing part of the zero-bias window. Fits to Fig. 3 give  $\text{CMW}_0 = 5.38$  and  $f_c = 179$  kHz for Stack A, corresponding to  $\tau = 0.90 \mu\text{s}$ , while Stack B relaxes much more slowly with  $\text{CMW}_0 = 3.6$  and  $f_c = 1.4$  kHz ( $\tau = 114 \mu\text{s}$ ). The full TMAP dataset shows that the differential loss and Cole–Cole representations return the same characteristic corner and do not require a distributed relaxation spectrum. That behavior is consistent with one dominant relaxation process rather than with a broad intrinsic ferroelectric response [9], [10].

Temperature provides an independent discriminator. At 10 K, Stack A still switches:  $2P_r$  decreases from 33.1 to  $20.3 \mu\text{C}/\text{cm}^2$  and  $E_c$  increases from 0.93 to  $1.17 \text{ MV}/\text{cm}$ , but the device remains ferroelectric [Fig. 5]. The capacitive readout changes more severely. The zero-bias CMW falls from 5.2 to about 2.2 at 1 kHz and then drops to the noise floor above 1.1 kHz, implying  $f_c \lesssim 1$  kHz at 10 K. Cooling therefore suppresses the capacitive-read bandwidth by more than two decades even though polarization reversal remains observable. That decoupling is difficult to reconcile with a read mechanism controlled only by the ferroelectric switching timescale.

The strongest argument comes from comparing the two stacks directly. Their low-frequency dielectric ceilings are nearly identical, so the ferroelectric layer presents essentially the same dielectric amplitude in both devices. Nevertheless, the relaxation times differ by  $126\times$  [Table I], and the stack containing the deliberate  $\text{WO}_x$  interface is the slower one. A compact Maxwell–Wagner description,

$$\tau \approx \frac{\varepsilon_i}{\sigma_i}, \quad (3)$$

assigns that timescale to the interfacial layer rather than to the ferroelectric bulk [11], [12]. The extracted interfacial areal capacitances differ only modestly ( $2.95$  versus  $2.34 \mu\text{F}/\text{cm}^2$ ), whereas the corresponding areal resistances differ by about  $157\times$ . In other words, the two stacks share nearly the same dielectric ceiling but not the same interface clock.

Taken together, the full plot set supports a simple reading rule for the reduced manuscript. The zero-bias CMW exists because polarization creates two stored states, but the rate at which those states remain capacitively distinguishable is controlled by interfacial charge relaxation. The metric reported at a single frequency is therefore not portable by itself. For conference presentation, the critical outputs are the low-frequency window, the fitted corner frequency, and the interface-sensitive comparison that shows why the same ferroelectric can yield very different NDRO read bandwidths.

### IV. CONCLUSION

The complete figure set and compressed two-page narrative support the same central conclusion as the full TMAP manuscript. The zero-bias CMW is not governed by the intrinsic ferroelectric timescale alone: its amplitude collapses with frequency while a substantial dielectric response remains, its relaxation follows a single-Debye process, and its bandwidth freezes on cooling much more strongly than polarization switching itself. Because two stacks with nearly identical low-frequency dielectric ceilings differ by more than two orders of magnitude in relaxation time, the usable NDRO read band is set by the interface through a Maxwell–Wagner mechanism. Zero-bias CMW values should therefore be reported together with stack details and measurement frequency.

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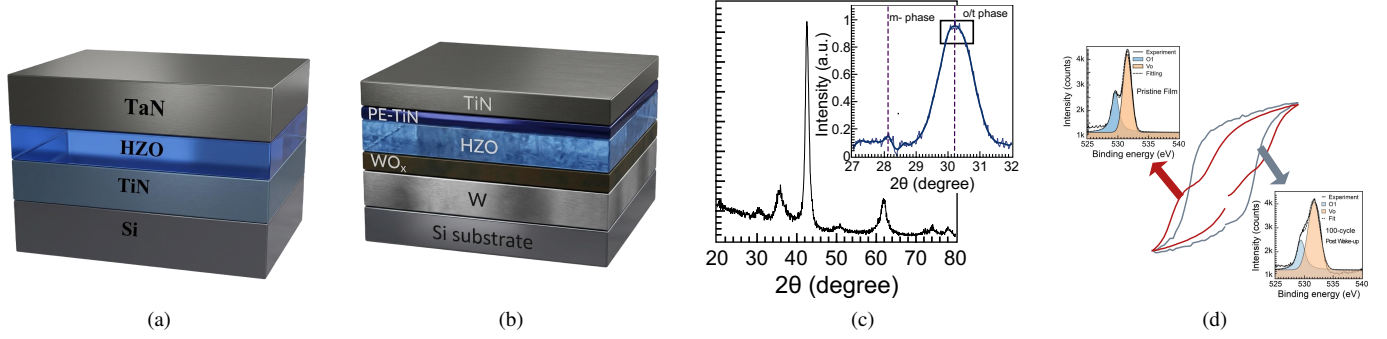


Fig. 1. Device and materials context from the TMat study. (a), (b) The two capacitor stacks differ in interface design, with Stack B including a deliberate  $\text{WO}_x$  layer. (c) GIXRD confirms ferroelectric-phase formation in the  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  films. (d) Polarization and interfacial-chemistry context motivate the interface-focused interpretation used in the reduced manuscript.

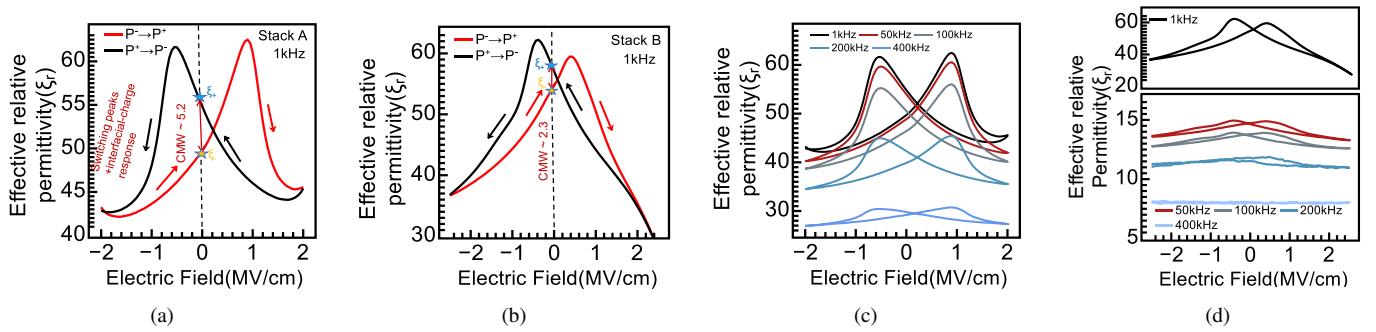


Fig. 2. Definition and frequency collapse of the zero-bias CMW. Stack A shows  $\text{CMW}_\xi \approx 5.2$  at 1 kHz, whereas Stack B shows  $\approx 2.3$  at a similar dielectric peak. With increasing frequency, the branch separation collapses while substantial permittivity remains, showing that the lost quantity is state discrimination rather than the entire dielectric response.

TABLE I  
KEY PARAMETERS RETAINED FOR THE REDUCED TMAT MANUSCRIPT

| Parameter              | Stack A                              | Stack B                        |
|------------------------|--------------------------------------|--------------------------------|
| Peak $\xi_r$ at 1 kHz  | 62.5                                 | 62.2                           |
| Ceiling $\xi_r(0)$     | $62.4 \pm 1.2$                       | $62.7 \pm 0.9$                 |
| CMW at 1 kHz           | 5.2                                  | 2.4                            |
| CMW <sub>0</sub> (fit) | 5.38                                 | 3.6                            |
| $f_c$                  | 179 kHz                              | 1.4 kHz                        |
| $\tau$                 | 0.90 $\mu$ s                         | 114 $\mu$ s                    |
| $E_{\text{cross}}$     | 0.221 MV/cm                          | 0.081 MV/cm                    |
| $c_{IL}$               | 2.95 $\mu$ F/cm <sup>2</sup>         | 2.34 $\mu$ F/cm <sup>2</sup>   |
| $R_{IL}$               | 0.31 $\Omega$ ·cm <sup>2</sup>       | 48.8 $\Omega$ ·cm <sup>2</sup> |
| $2P_r$ at 300 K / 10 K | 33.1 / 20.3 $\mu$ C/cm <sup>2</sup>  |                                |
| CMW at 300 K / 10 K    | 5.2 / 2.2 (Stack A)                  |                                |
| $f_c$ at 300 K / 10 K  | 179 kHz / $\lesssim$ 1 kHz (Stack A) |                                |

